

BROOKE A. MENDEZ

Senior Data Scientist | Machine Learning Engineer | AI/ML Developer

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PROFESSIONAL SUMMARY

Senior Data Scientist and Machine Learning Engineer with 8+ years of combined experience building data products, predictive models, Generative AI applications, and production analytics systems. Strong background in Python, SQL, Spark, Databricks, AWS, Scikit-learn, TensorFlow, NLP, LLM applications, and retrieval-augmented generation (RAG). Experienced across the full ML lifecycle—from data preparation and feature engineering through evaluation, deployment, monitoring, and stakeholder-facing analytics—with hands-on work improving retrieval quality, LLM response quality, latency, and model performance.

CORE SKILLS

Programming & Analytics: Python • SQL • R • Pandas • NumPy • SciPy • Jupyter

Machine Learning: Scikit-learn • TensorFlow • PyTorch • XGBoost • Regression • Classification • Clustering • Random Forest • Decision Trees • Neural Networks • Feature Engineering

Generative AI & NLP: LLMs • RAG • Embeddings • Vector Databases • Semantic Search • Prompt Engineering • OpenAI APIs • LangChain • Question Answering • Document Summarization

AI & Model Evaluation: LLM Evaluation • Retrieval Evaluation • Response Quality Assessment • Semantic Relevance • Accuracy/Latency Benchmarking • Cross-Validation • Precision • Recall • F1 • ROC-AUC • RMSE • MAE

Data Engineering: ETL/ELT • Data Wrangling • Data Cleaning • Apache Spark • Airflow • Databricks • Snowflake • BigQuery • Data Pipelines

Cloud & MLOps: AWS • Docker • Kubernetes • REST APIs • FastAPI • Model Deployment • Model Monitoring

Visualization & Databases: Tableau • Power BI • Matplotlib • Seaborn • Plotly • PostgreSQL • MySQL • MongoDB • Redis

Statistics: Probability • Hypothesis Testing • A/B Testing • Time Series • Hyperparameter Tuning

PROFESSIONAL EXPERIENCE

Senior Data Scientist

Nov 2024 – Present

Allwyn Corporation — Arlington, VA | Remote

Project: Generative AI Bot for Citizen Services — Allwyn Digital Modernization Platform

Url: allwyncorp.com

- Developed LLM-powered citizen-service applications using RAG and LangChain, reducing manual knowledge-retrieval effort by 8% and improving response time by 6%.
- Built semantic-search and NLP pipelines for question answering, document summarization, and conversational AI using embedding models and vector-search workflows.
- Designed Python, Spark, Databricks, and AWS pipelines processing approximately 500 GB of structured and unstructured data per day for downstream analytics and AI workflows.
- Evaluated and tuned retrieval and LLM application quality through prompt iteration, vector-search optimization, semantic relevance checks, and evaluation frameworks, achieving 4.5% lower latency and 12% higher measured accuracy.
- Partnered with engineering and product teams to deploy production-grade AI services and improve reliability, automation, and end-user support workflows.

Senior Data Scientist

Mar 2020 – Dec 2022

Cardinality.ai — Gaithersburg, MD | Remote

Project: Embedded AI & Intelligence Platform

Url: prnewswire.com

- Developed predictive analytics and machine-learning models for government AI platforms using Python, SQL, and Scikit-learn, improving prediction accuracy by 7% over baseline models.
- Built scalable ETL pipelines and automated data workflows for large structured and unstructured datasets supporting analytics and model development.
- Applied NLP, statistical analysis, feature engineering, cross-validation, and model evaluation techniques to improve operational intelligence and citizen-outcome predictions.
- Designed Tableau and Power BI dashboards for KPI tracking, real-time analytics, and executive reporting.
- Deployed cloud-based ML solutions using AWS, Docker, Airflow, Snowflake, and Kubernetes to improve portability, scalability, and production reliability.

Machine Learning Engineer

Jan 2018 – Feb 2020

[Hypergiant Industries — Austin, TX | Remote](#)

Project: Hypergiant CommandCenter Platform

Url: [hypergiant.com](#)

- Developed predictive analytics and machine-learning models for enterprise AI applications using Python and Scikit-learn.
- Analyzed large-scale structured and unstructured datasets, performed feature engineering, and translated model outputs into operational business insights.
- Built automated ETL pipelines and data workflows using SQL, Airflow, and cloud analytics tools, reducing manual reporting effort by 3%.
- Developed Snowflake data models and AWS ingestion pipelines using S3 and Redshift to support scalable business-intelligence reporting.
- Managed Docker-based data services and Kubernetes workloads for analytics applications and created dashboards for real-time operational monitoring.

Data Analyst

Jun 2015 – Dec 2017

[Airtable — San Francisco, CA | Remote](#)

Project: Airtable Product Usage & Feature Adoption Analytics

Url: [airtable.com](#)

- Analyzed user-interaction data for Gallery View and Kanban View feature launches to identify product-adoption and engagement trends.
- Built SQL dashboards to monitor feature adoption, retention, engagement, and workflow-usage metrics.
- Performed exploratory data analysis using Python and Tableau and delivered KPI reports and analytical insights to product and business teams.

CAREER BREAK

Jan 2023 – Oct 2024 | Took time away from full-time work before returning to data science in late 2025.

SELECTED AI/ML PROJECT HIGHLIGHTS

- Generative AI Citizen Services: RAG-based question answering, summarization, semantic search, prompt optimization, and LLM/retrieval evaluation for production support workflows.
- Government Predictive Intelligence: Python/SQL machine-learning workflows combining structured and unstructured data, statistical analysis, NLP, feature engineering, model evaluation, and executive BI reporting.
- Enterprise Analytics Platform: ML-driven analytics, Snowflake data models, AWS S3/ Redshift ingestion, Airflow orchestration, containerized services, and Kubernetes-based deployment

EDUCATION

Master of Science in Data Science

Sep 2015 – Mar 2018

[Florida Institute of Technology — Melbourne, FL](#)

Relevant Coursework: Machine Learning • Data Mining • Statistics • Database Systems • Artificial Intelligence • Data Analytics • Predictive Modeling • Software Engineering

Bachelor of Science in Information Systems

Aug 2011 – May 2015

[Florida Institute of Technology — Melbourne, FL](#)

Relevant Coursework: Database Systems • Programming • Statistics • Systems Analysis • Business Information Systems • Data Management • Web Development